

FinTech and AI in Finance

Class 5: Interpretable Machine Learning and Portfolio Decisions

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Interpreting Characteristics and Forecasts

OLS Coefficients: Size and Statistical Significance

A coefficient measures how the forecast changes when one input rises, holding the others fixed.

Standardized predictor	Coefficient	Standard error	<i>p</i> -value
Momentum	+0.30	0.10	0.003
Volatility	−0.20	0.25	0.42

- ▶ Momentum: +0.30 percentage points per standard deviation; significant at 5%.
- ▶ Volatility: negative estimate; not significant at 5%.

Coefficients and standard errors are percentage points per training-sample standard deviation.

Linear Coefficients: Calculating Forecast Contributions

Suppose a fitted weekly excess-return rule uses standardized momentum and volatility:

$$\hat{r}^e = 0.10\% + 0.30\% z_{\text{momentum}} - 0.20\% z_{\text{volatility}}.$$

For a stock with momentum 1 and volatility 0.5:

Term	Calculation	Contribution
Intercept	0.10%	+0.10%
Momentum	$0.30\% \times 1$	+0.30%
Volatility	$-0.20\% \times 0.5$	-0.10%
Forecast	sum of contributions	+ 0.30%

OLS still uses volatility despite its insignificant slope.

LASSO Coefficients: Selected or Set to Zero

Suppose validation selects a LASSO penalty for the same standardized predictors:

Predictor	OLS coefficient	LASSO coefficient	LASSO selection
Momentum	+0.30	+0.24	retained
Volatility	−0.20	0	excluded

$$\hat{r}^e = 0.10\% + 0.24\% z_{\text{momentum}}.$$

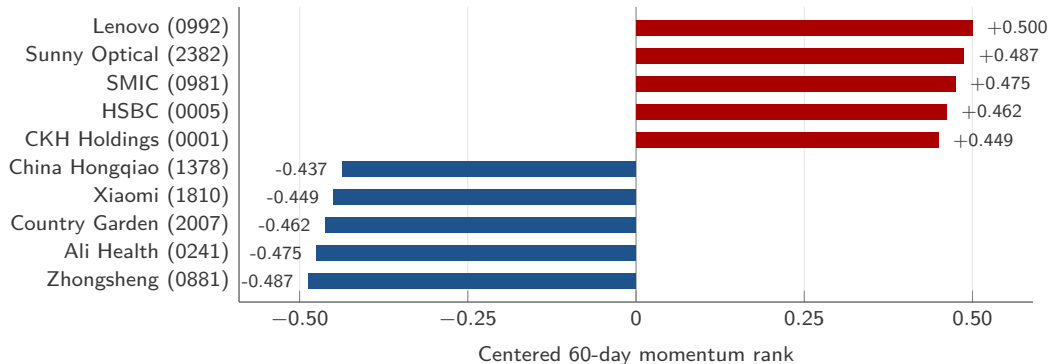
At $z_{\text{momentum}} = 1$, the forecast is 0.34%. Volatility contributes zero at this penalty.

LASSO selection concerns prediction, not statistical significance.

Coefficients are percentage points per training-sample standard deviation.

A Weekly Momentum Ranking

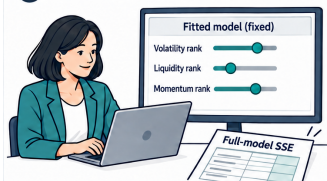
Week ending 26 Jun 2026: five highest and five lowest of 79 available ranks.



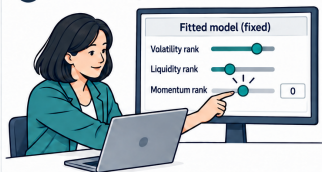
Pre-week ranks describe an input, not a forecast or an importance score.

Drop-to-Zero Importance: Which Inputs Does the Model Use?

1 Record full-model fit



2 Set one input to zero



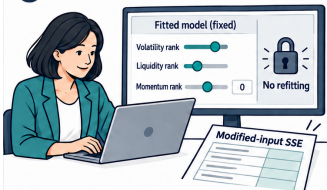
Keep the model and observations fixed.

Set one centered characteristic rank to zero and compare the fitted error:

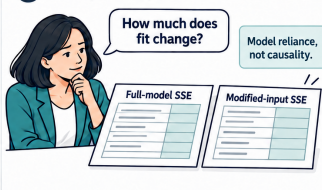
$$\Delta R_j^2 = \frac{\text{SSE}_{j=0} - \text{SSE}_{\text{full}}}{\sum_{i,t} (r_{i,t}^e)^2}.$$

A negative score means zeroing that input improves fit. Do not refit.

3 Run the SAME fitted model



4 Compare squared errors



Method: Gu, Kelly, and Xiu (2020), RFS, Sections 1.9 and 2.3.

Drop-to-Zero Importance: A Worked Calculation

On the same training observations, suppose the zero-forecast benchmark has squared error $\sum(r_{i,t}^e)^2 = 20$.

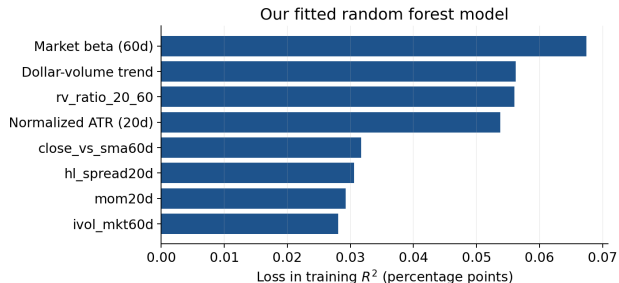
Predictions	Sum of squared errors	Fitted R^2
Original fitted model	9.6	$1 - 9.6/20 = 0.52$
Set momentum rank to zero	11.2	$1 - 11.2/20 = 0.44$

$$\Delta R_{\text{momentum}}^2 = 0.52 - 0.44 = 0.08.$$

Momentum's score is **8 percentage points**. Repeat without refitting.

These are raw importance scores, not shares of total importance.

Characteristic Importance in Our Fitted Model



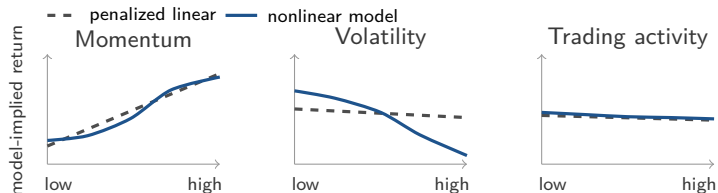
Set each centered rank to zero and compare fitted R^2 .

- ▶ Larger score: greater reliance in this fit.
- ▶ Correlated substitutes can reduce a score.
- ▶ The score has no direction.

Largest scores: market beta, dollar-volume trend, and short-to-long volatility ratio. These are training-fit diagnostics, not OOS accuracy.

GKX-style fixed-model diagnostic; positive scores; zero excess-return benchmark.

Response Curves: How Predictions Change



A response curve shows whether an important input raises or lowers the forecast, and where its effect changes.

Vary that input while fixing the others at reference values. Unlike an importance score, the curve shows direction and thresholds. It describes the fitted rule, not a causal effect.

Method reference: Gu, Kelly, and Xiu (2020), *RFS*, Section 1.9.

Response Curves: Calculating One Curve

Suppose a fitted rule uses centered momentum and volatility ranks:

$$\hat{r}^e = 0.10\% + 0.40\% \max(0, m - v).$$

Fix volatility at its middle rank, $v = 0$, and vary momentum:

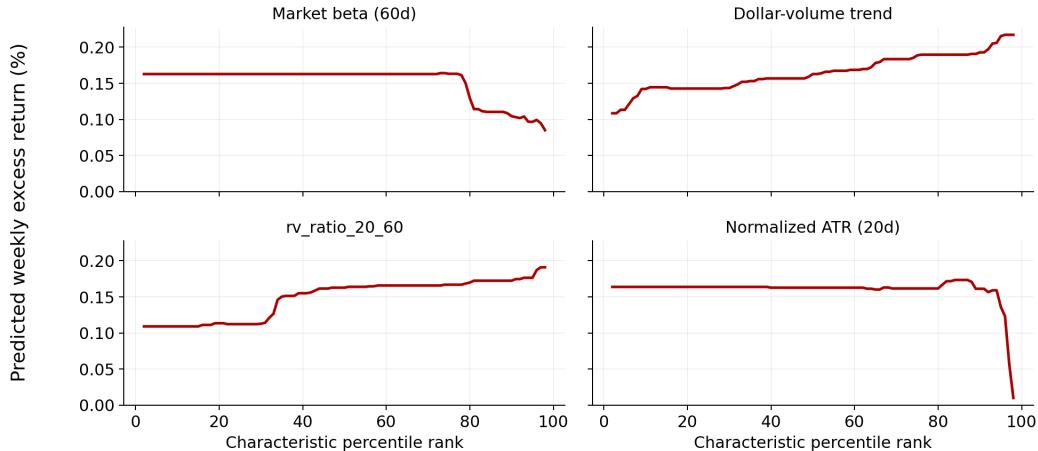
Momentum rank m	-0.5	0	+0.5
$\max(0, m - 0)$	0	0	0.5
Predicted excess return	0.10%	0.10%	0.30%

Plot the last row against m : a flat segment followed by an upward slope. For our empirical curves, hold every other centered rank at zero.

This curve holds other inputs at reference values instead of averaging across stocks.

Response Curves in Our Fitted Stock Model

Random forest: vary one characteristic, hold other ranks at zero



Cross-Sectional Portfolio Decisions

A Top-Bottom Spread Tests a Ranking

Equal-weight the highest- and lowest-forecast groups, H and L :

$$r_t^{H-L} = \frac{1}{n_H} \sum_{i \in H} r_{i,t} - \frac{1}{n_L} \sum_{i \in L} r_{i,t}.$$

Long exposure = 1, short exposure = -1 : net exposure = 0, gross exposure = 2.

$$r_{p,t}^{\text{gross}=1} = \frac{1}{2} r_t^{H-L}.$$

Interpretation: tests the forecast ordering at the extremes. Dollar neutrality does not imply beta or sector neutrality.

The Class 3 spread table uses unit exposure per side; later weight examples use gross exposure one.

Replace the Characteristic with a Forecast

	Characteristic sort	Forecast sort
Ranking variable	One chosen input	The fitted return forecast
Information used	A single characteristic	Inputs combined by the fitted model
Formation rule	Same dates, holding weeks, and portfolio groups	

MSE evaluates the numerical forecast for every stock-week. A top-bottom spread evaluates the ordering of stocks at the extremes.

A model can rank stocks usefully without producing accurate return magnitudes.

Class 3's Six Forecasts, Sorted

Weekly top-minus-bottom quintile spread over the frozen 52-week Hang Seng Index test period.

Model	Mean spread	t	Hit rate	First half	Second half
OLS	+0.32%	1.03	55.8%	-0.07%	+0.71%
Ridge	+0.48%	1.46	53.8%	+0.13%	+0.83%
LASSO	<i>constant forecast: all slopes are zero</i>				
Elastic Net	<i>constant forecast: all slopes are zero</i>				
PCR	+0.46%	1.41	55.8%	-0.33%	+1.26%
PLS	+0.42%	1.26	57.7%	-0.35%	+1.19%

The spread is a ranking test, not a forecast-loss test.

Keep in mind

LASSO and Elastic Net are constant forecasts. PCR is the validation choice; Ridge has the highest observed spread.

Long-Only and Long-Short Answer Different Mandates

	Fully invested long-only	Dollar-neutral long-short
Weights	$w_i \geq 0, \sum_i w_i = 1$	$\sum_i w_i = 0$; shorts allowed
Exposure	Full capital invested; beta varies	Zero net dollars; beta can be nonzero
Evidence	Return relative to the benchmark	Spread return, risk, and factor exposures
Constraints	Concentration and position limits	Borrow, margin, short costs, and recalls

One ranking can support either mandate; evaluate it under the relevant constraints and costs.

Three Ways to Turn One Forecast into Weights

Demmean forecasts within a formation week: $s_i = \widehat{r}_i^e - \overline{\widehat{r}^e}$.

Rule	Weights at gross exposure one	Uses
Quintile spread	$w_i = 1/(2n_H)$ in H ; $-1/(2n_L)$ in L ; 0 outside these groups	Ordering at the extremes
Forecast weighted	$w_i = s_i / \sum_j s_j $	Ordering and magnitude
Long-only tilt	$u_i = \max(0, b_i + \eta s_i)$; $w_i = u_i / \sum_j u_j$	Benchmark b_i and forecast tilt

b_i : benchmark weights; $\eta \geq 0$: tilt strength. Enforce position limits.

If all $s_i = 0$, skip both long-short rules and set $w_i = 0$ to avoid 0/0. For the long-only tilt, keep the benchmark weights: $w_i = b_i$.

Put the Three Rules on the Same Footing

Exposure measures:

$$G = \sum_i |w_i| \quad (\text{gross}), \quad N = \sum_i w_i \quad (\text{net}).$$

At $G = 1$, the spreads have $N = 0$; the fully invested long-only tilt has $N = 1$. Equal gross exposure does not equalize risk.

If volatility targeting is permitted, scale each strategy p separately:

$$k_{p,t-1} = \frac{\sigma_{\text{target}}}{\widehat{\sigma}_{p,t-1}}, \quad w_{p,t-1}^{\text{scaled}} = k_{p,t-1} w_{p,t-1}.$$

Use lagged risk estimates at the same frequency. Different strategy risks imply different k ; scaling changes gross exposure and must respect leverage and cash constraints.

Worked Example: Forecast-Weighted Positions

Four demeaned forecasts are $s = (0.3, 0.1, -0.1, -0.3)\%$. With gross exposure one,

$$w_i = \frac{s_i}{\sum_j |s_j|}, \quad \sum_j |s_j| = 0.8\%.$$

Stock	A	B	C	D
Weight	37.5%	12.5%	-12.5%	-37.5%
Realized return	2%	1%	-1%	-2%

$$r_p = 0.375(2\%) + 0.125(1\%) - 0.125(-1\%) - 0.375(-2\%) = 1.75\%.$$

Net exposure is zero; gross exposure is 100%.

Calculation Question: Forecast-Weighted Positions

Suppose four stocks have the following forecasts and subsequent returns:

Stock	A	B	C	D
Forecast excess return	0.8%	0.4%	−0.2%	−0.6%
Next-week realized return	2%	−1%	1%	−3%

Use $s_i = \hat{r}_i^e - \overline{\hat{r}^e}$ and $w_i = s_i / \sum_j |s_j|$.

1. Calculate the mean forecast, the demeaned forecasts, and the four weights.
2. Calculate gross exposure and net exposure.
3. Calculate the spread return $\sum_i w_i r_i$, ignoring transaction costs.

Cumulative Wealth Shows the Path, Not Just the Average

$$\text{Cumulative wealth is } W_T = \prod_{t=1}^T (1 + r_{p,t}).$$



HSI momentum terciles. High-Low uses one unit per side and compounds weekly spreads.

Return and Risk at a Common Frequency

Let $K = 12$ for monthly returns and $K = 52$ for weekly returns.

Statistic	Definition
Annualized arithmetic mean	$K \bar{r}_p$
Annualized volatility	$\sqrt{K} s(r_p)$
Annualized Sharpe ratio	$\sqrt{K} (\bar{r}_p - r_f) / s(r_p - r_f)$
Compound annual return	$\left[\prod_{t=1}^T (1 + r_{p,t}) \right]^{K/T} - 1$

Each example states the cash convention. Square-root annualization ignores serial dependence.

Worked Example: Average Return and Compounded Wealth

A portfolio earns +10% in the first month and −10% in the second.

$$\bar{r} = (10\% - 10\%)/2 = 0, \quad W_2 = 1.10 \times 0.90 = 0.99.$$

The average monthly return is zero, but the investor loses 1%.

Arithmetic mean describes average periodic return; compounded wealth describes capital growth. Sharpe uses mean excess return, not compound annual return.

Multiple-Choice Question: Annualized Sharpe Ratio

A weekly strategy has an average return of 0.30%, a constant weekly risk-free return of 0.05%, and return volatility of 1.80%. Ignore serial correlation and use

$$\text{Sharpe} = \frac{\bar{R}_p - \bar{R}_f}{s(R_p)} \sqrt{52}.$$

What is the annualized Sharpe ratio?

- A. 0.14
- B. 1.00
- C. 1.20
- D. 7.22

Discuss

Identify the errors behind using total return instead of excess return, or multiplying a weekly Sharpe ratio by 52 instead of $\sqrt{52}$.

Multiple-Choice Question: Drop-to-Zero Importance

On the same observations, the zero-forecast benchmark has squared error 200. A fitted model has squared error 80. Setting one centered characteristic to zero raises the error to 100, with the model otherwise unchanged.

What is the reduction in fitted R^2 ?

- A) 0.10, or 10 percentage points.
- B) 0.20, or 20 percentage points.
- C) 0.40, or 40 percentage points.
- D) 0.50, or 50 percentage points.

Multiple-Choice Question: Portfolio Weights and Exposure

Suppose a portfolio has weights $(0.30, 0.20, -0.25, -0.25)$. The four stocks return $(2\%, 4\%, -2\%, 0\%)$ over the next period. Ignore costs.

Which pair gives the portfolio's gross exposure and return?

- A) Gross exposure 0 and return 1.90%.
- B) Gross exposure 1.00 and return 1.90%.
- C) Gross exposure 1.00 and return 0.90%.
- D) Gross exposure 0 and return 0.90%.

Market Timing and Risk

Switching from Stock Selection to Market Timing

Design	Time-series timing	Cross-sectional selection
Question	how much market exposure now?	which assets should receive more weight?
Rows	dates for one asset or market	assets observed at each date
Target	next-period market log return	next-period excess return of each asset
Output	exposure through time	relative ranking across assets

Buy-and-Hold as the Investment Benchmark

Let $\ell_{m,t}$ be the monthly log market return. Its simple total return is $r_{m,t} = \exp(\ell_{m,t}) - 1$.

Buy-and-hold uses

$$w_{t-1}^{\text{BH}} = 1, \quad r_{p,t}^{\text{BH}} = \exp(\ell_{m,t}) - 1.$$

- ▶ It requires no return forecast and stays fully invested every test month.
- ▶ Class 2's three-predictor OLS timing rule and buy-and-hold use the same market proxy and 60 test months.
- ▶ No transaction cost is included for either strategy.
- ▶ Cash earns zero in this simple classroom exercise.

One Forecast Sets Next Month's Market Weight

At month-end $t - 1$, compare the forecast log return with its training-sample mean:

$$w_{t-1} = \mathbb{I}\{\hat{\ell}_{m,t} > \bar{\ell}_{\text{train}}\}, \quad w_{t-1} \in \{0, 1\}.$$

Convert the realized log return to a simple return before applying the weight:

$$r_{p,t}^{\text{timing}} = w_{t-1} [\exp(\ell_{m,t}) - 1].$$

- ▶ $w_{t-1} = 1$ holds the market proxy during month t ; zero holds cash.
- ▶ The exercise compares market exposure and abstracts from transaction costs.
- ▶ Fix the weight before observing month t .
- ▶ Fit and evaluate in log returns; invest using simple returns.
- ▶ The threshold is not the MA(12) OOS benchmark.

One Forecast, Two Benchmarks, Three Markets

Market	OOS R^2 vs MA(12)	Hit rate / positive base rate
U.S.: SPY	6.7%	63.3% / 63.3%
HKSAR: Hang Seng	4.4%	41.7% / 46.7%
Chinese Mainland: CSI 300	4.3%	51.7% / 48.3%

Market	BH ann. mean	Timing ann. mean	BH Sharpe	Timing Sharpe	Timing exposure
U.S.: SPY	13.8%	7.1%	0.87	0.51	75.0%
HKSAR: Hang Seng	-1.7%	-4.7%	-0.07	-0.24	41.7%
Chinese Mainland: CSI 300	0.6%	1.8%	0.03	0.11	83.3%

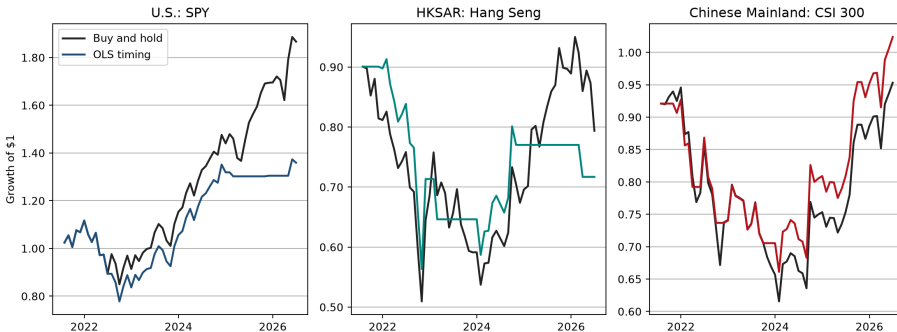
167 pre-test months; 60-month test (Jul. 2021–Jun. 2026). Fit/OOS loss: log returns. Portfolio arithmetic: simple returns; cash and costs: zero.

Keep in mind

OLS beats MA(12) on MSE, but timing trails buy-and-hold in the U.S. and HKSAR.

One Timing Rule Creates Three Investment Paths

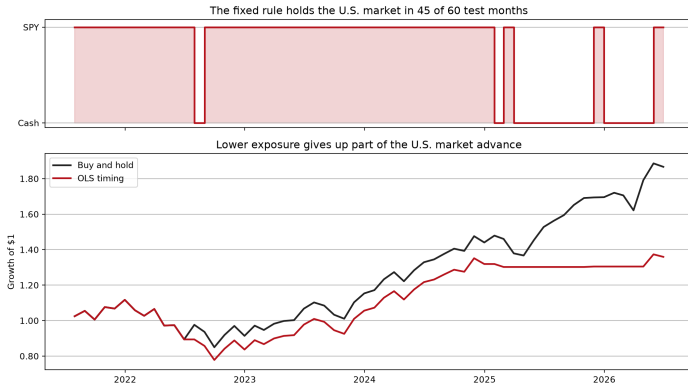
A fixed long-or-cash rule creates three different investment paths



Keep in mind

The same long-or-cash mapping can help, hurt, or mostly change the path. The exercise changes market exposure; it does not select individual stocks.

U.S. Timing Still Trails Buy-and-Hold



Keep in mind

Annualized mean is 7.1% versus 13.8% for buy-and-hold. The following examples examine how continuous weights change the return and risk of the investment.

Hang Seng and CSI 300: The Same Rule, Opposite Outcomes

Market	Months held	Mean return when held	Months in cash	Mean return when in cash
HKSAR: Hang Seng	25	-0.9%	35	+0.4%
Chinese Mainland: CSI 300	50	+0.2%	10	-0.6%

Terminal wealth of one dollar over the 60 test months: Hang Seng buy-and-hold 0.79 versus timing 0.72; CSI 300 buy-and-hold 0.95 versus timing 1.02.

Keep in mind

CSI 300 avoids months averaging -0.6% and finishes ahead. Hang Seng holds months averaging -0.9% but skips months averaging +0.4% and finishes behind. Which months are held matters more than how often.

Position Sizing with the Current Market Excess-Return Forecasts

Use the validation-selected U.S. LASSO and Gradient boosting forecasts, with lagged 12-month SPY volatility.

Choose the market weight to balance expected return and variance:

$$\max_{0 \leq w \leq 1} \left\{ w \hat{\mu}_t - \frac{\gamma}{2} w^2 \hat{\sigma}_t^2 \right\} \implies w_{t-1}^* = \max \left(0, \min \left(1, \frac{\hat{\mu}_t}{\gamma \hat{\sigma}_t^2} \right) \right).$$

$\hat{\mu}_t$: monthly expected excess return; $\hat{\sigma}_t^2$: monthly variance; $\gamma > 0$: risk aversion. All inputs are known at $t - 1$.

Test: July 2021–June 2026. The uninvested balance earns the period-matched U.S. 3-month Treasury return, so $\hat{\mu}_t = \hat{r}_{m,t}^e$.

Worked Example: Forecast and Risk Determine the Weight

For August 2023, boosting forecasts 0.337% for SPY; lagged annualized volatility is 19.09%. Set $\gamma = 5$ and $0 \leq w \leq 1$.

$$w_{t-1} = \max\left(0, \min\left(1, \frac{\hat{r}_{m,t}^e}{\gamma \hat{\sigma}_{\text{monthly},t}^2}\right)\right).$$

Convert annualized volatility to monthly variance:

$$w = \frac{0.337\%}{5 (19.09\% / \sqrt{12})^2} \approx 0.222 = 22.2\%.$$

Risk-free return: 0.45%; SPY return: -1.63%; portfolio return: -0.01%.

Uninvested capital earns the risk-free return; costs are zero.

Continuous Weights Change Return and Risk

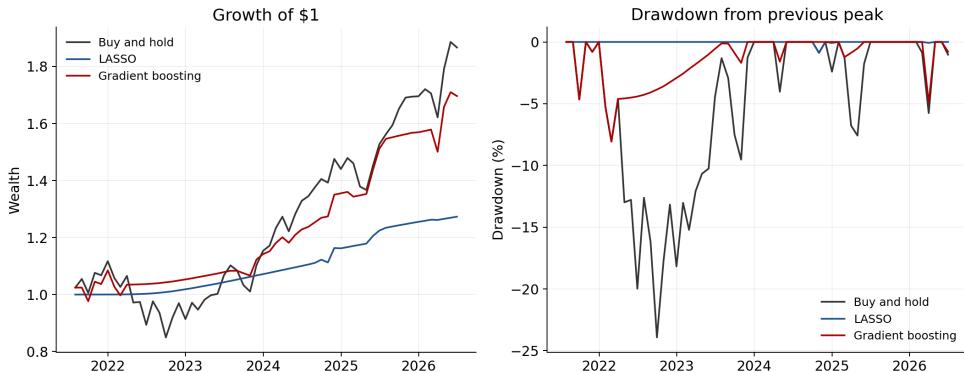
Strategy	Compound ann. return	Ann. vol.	Sharpe	Turnover
Buy and hold	13.3%	15.9%	0.65	0.0%
LASSO	4.9%	2.3%	0.57	4.6%
Gradient boosting	11.1%	9.1%	0.81	23.2%

Compound annual growth uses total returns; Sharpe uses monthly excess returns.

Lower exposure can reduce both drawdowns and average returns.

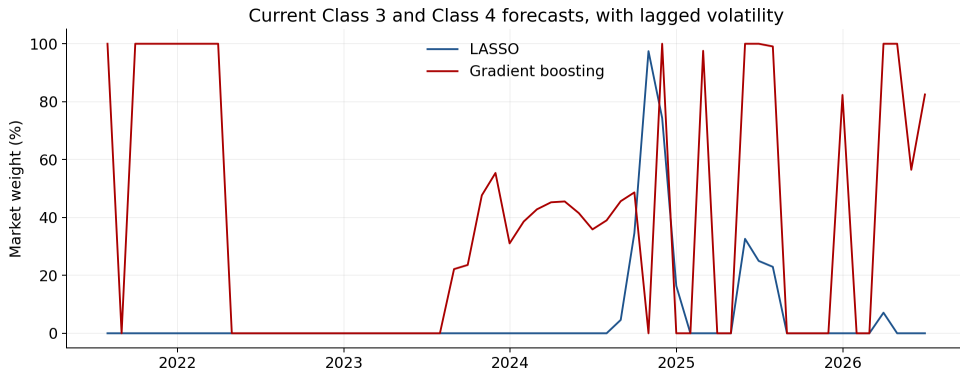
U.S. test sample: Jul. 2021–Jun. 2026. Cash earns the risk-free return; costs are zero.

Portfolio Outcomes over the Same Market Path



All three strategies experience the same SPY returns. Their different weights produce different compounded wealth and drawdown paths.

Forecasts and Volatility Jointly Determine Weights



For a positive excess-return forecast, greater estimated variance reduces the market position. Negative forecasts produce a zero market weight and leave capital at the risk-free rate.

Transaction Costs: A 10 bp Example

Assume a cost of 10 basis points on the **value of each purchase or sale**:

$$10 \text{ bp} = 0.10\% = 0.001.$$

Weights immediately before and after rebalancing, before deducting costs:

Stock	Before	After	Trade as % of portfolio value
A	50%	70%	Buy 20%
B	50%	30%	Sell 20%

$$\text{Cost} = 0.001 \times (0.20 + 0.20) = 0.0004 = \mathbf{0.04\%}.$$

For a HK\$1 million portfolio, this rebalance costs **HK\$400**, or 4 bp of portfolio value.

Gross and Net Sharpe Ratios

A hypothetical strategy has a mean weekly **gross excess return of 0.20%** and weekly excess-return volatility of **2.00%**.

Assume purchases plus sales total 40% of portfolio value every week. At 10 bp on traded value, the weekly cost is **0.04%**.

Before costs

$$SR_{\text{gross}} = \frac{0.20\%}{2.00\%} \sqrt{52} = \mathbf{0.72}.$$

After costs

$$SR_{\text{net}} = \frac{0.20\% - 0.04\%}{2.00\%} \sqrt{52} = \mathbf{0.58}.$$

The same cost every week lowers the mean excess return to 0.16%. Subtracting a constant leaves volatility unchanged at 2.00%.

Calculation Question: Trading Costs and Sharpe Ratios

A strategy buys 30% and sells 30% of portfolio value each week. Each purchase or sale costs 10 bp of traded value.

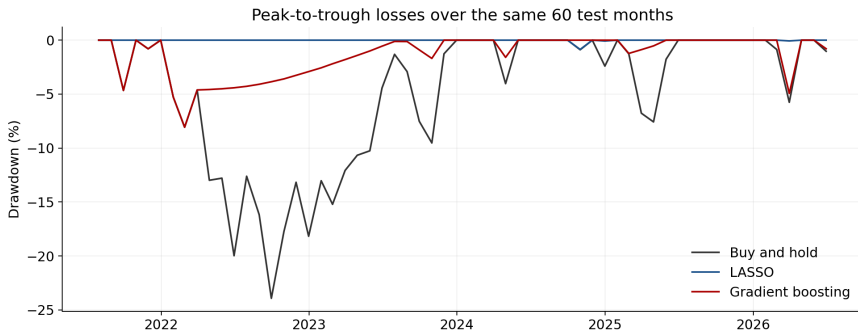
Its gross weekly return has mean 0.30% and standard deviation 1.80%. The weekly risk-free return is a constant 0.05%. The same percentage cost is deducted each week, so volatility is unchanged.

Use $SR_{\text{annual}} = \sqrt{52} (\bar{r}_p - r_f) / s(r_p)$.

1. Calculate weekly trading costs as a percentage of portfolio value.
2. Calculate the mean net return and the mean net excess return.
3. Calculate annualized Sharpe ratios before and after costs.

Maximum Drawdown Measures the Largest Peak-to-Trough Loss

$$\text{Drawdown}_t = W_t / \max(1, W_1, \dots, W_t) - 1.$$



The maximum drawdown is the most negative point on each line. Include the initial wealth of one when identifying the previous peak.

Alpha and Beta

Two People behind the Portfolio Framework



Harry Markowitz

1990 laureate

Portfolio selection

Mean–variance diversification and the efficient frontier.



William Sharpe

1990 laureate

CAPM and beta

Tangency, the market portfolio, and performance benchmarks.

A note on the Nobel prizes

The 1990 prize was shared with Merton Miller.

Sources: Nobel Prize facts for [Markowitz](#) and [Sharpe](#). Portraits are AI-generated editorial illustrations, not photographs.

Alpha and Beta Separate Performance from Market Exposure

Estimate a CAPM-style performance regression:

$$r_{p,t} - r_{f,t} = \alpha + \beta_M(r_{m,t} - r_{f,t}) + \varepsilon_t.$$

Beta: sensitivity to market excess return. If $\beta_M = 1.10$, a one-percentage-point market move changes fitted excess return by 1.10 points.

Alpha: average excess return beyond the fitted market contribution:

$$\hat{\alpha} = \overline{r_p - r_f} - \hat{\beta}_M \overline{r_m - r_f}.$$

OLS residuals average zero when an intercept is included; alpha need not be zero. Use the local market benchmark.

Beta Moves with the Market; Alpha Shifts the Fitted Return

Suppose a strategy has monthly $\alpha = 0.2\%$ and $\beta_M = 1.10$. Its fitted excess return is

$$\widehat{r_p - r_f} = 0.2\% + 1.10(MKT - RF).$$

Market excess return	Beta contribution	Alpha	Fitted portfolio excess return
-2.0%	-2.2%	+0.2%	-2.0%
0.0%	0.0%	+0.2%	+0.2%
2.0%	+2.2%	+0.2%	+2.4%

Add r_f for fitted total return. Realized return also includes the residual ε_t .

Keep in mind

Beta changes the response to market moves. Alpha is the intercept, estimated as an average across months. Neither one is a one-month residual.

Empirical regressions use the period-matched risk-free return.

Calculation Question: What Is the CAPM Benchmark Return?

In one week, $R_f = 0.10\%$ and $MKT - RF = 1.00\%$. A portfolio has a market beta of 1.10 and a realized return of 1.40%.

$$R^{\text{CAPM}} = R_f + \beta_M(MKT - RF), \quad \text{abnormal return} = R_p - R^{\text{CAPM}}.$$

1. Calculate the CAPM benchmark return and show the substitution.
2. Calculate the portfolio's one-week abnormal return.
3. Explain why one positive week does not establish a positive CAPM alpha.

Show your reasoning

Separate the benchmark calculation from the interpretation of performance.

CAPM Alpha Must Use a Local Market Benchmark

Local market proxy	Annualized α	$t(\alpha)$	β_M	R^2
U.S.: S&P 500 (adjusted SPY)	-4.05%	-1.23	0.74	72.6%
HKSAR: Hang Seng	-4.77%	-0.92	0.67	66.9%
Chinese mainland: CSI 300	+1.10%	0.40	0.87	87.4%

Each row evaluates the long-or-cash OLS timing strategy against its local market. CAPM uses period-matched excess returns; costs are zero. Test: 2021-07–2026-06.

Keep in mind

Chinese mainland has the only positive alpha estimate, but none is statistically distinguishable from zero.

Interpreting Estimated CAPM Alpha

- ▶ Alpha is average excess return after allowing for estimated local-market beta.
- ▶ Its standard error describes how precisely that average is estimated.
- ▶ A different benchmark or omitted exposure can change the estimated alpha.

Our three timing strategies have alpha estimates that are not statistically distinguishable from zero in the 60-month test.

Alpha and Beta for Five Large HSI Constituents

Company	Aug. 2026 weight	Ann. mean	Ann. α	$t(\alpha)$	β_{HSI}	R^2
HSBC (0005.HK)	8.91%	+39.0%	+32.4%	3.61	0.53	29.0%
Tencent (0700.HK)	7.67%	+13.2%	+4.0%	0.37	1.19	68.6%
Alibaba (9988.HK)	7.11%	+13.0%	+2.2%	0.13	1.59	63.5%
CCB (0939.HK)	5.47%	+27.1%	+20.3%	2.19	0.58	37.8%
AIA (1299.HK)	4.79%	+4.9%	-4.1%	-0.32	1.15	52.9%

Keep in mind

Beta differs sharply across firms; alpha depends on the sample and benchmark.

HSI weights: 31 Aug. 2026. OLS: 156 weekly observations, Jul. 2023–Jun. 2026; Newey–West(4) t -statistics.

Assignment Data and Portfolio Rules

Period	Weeks	Stock-week rows	Next-week returns
Training	1–78	6,240	visible
Public validation	79–104	2,080	visible
Private evaluation	105–260	12,480	hidden

80 stocks, 17 simulated signals. Target: $r - r_f$. OOS R^2 benchmark: zero.

Portfolio rule	Submission requirement
Position limits	Long-only, fully invested, and no stock above 10%
Evaluation period	Rebalance weekly for the 156 private weeks
Investment benchmark	Weekly rebalanced equal weight across all 80 stocks
Trading costs	Ignored. Turnover may be reported as a diagnostic

The ML Investment Assignment

Course weight	Team	Release	Due
20%	Up to three students	After Class 4	15 Oct 2026, 2:00 p.m. (HKT)

Submit

- ▶ PDF report: max. 8 pages + optional appendix.
- ▶ Weight file: 156 rows, 81 columns.
- ▶ Reproducible R/Python code; ZIP max. 10 MB.

Assessment

Data and benchmark	10
Model comparison	35
Interpretation	15
Portfolio evidence	15
Reproducibility	5
Private performance	20
Total	100

Before submission

Run the checker; compare OLS, LASSO, and random forest or boosting. Disclose AI assistance.